Offer Decision Rules

Smart Offer Engine · every offer, from the AI or the rules, is checked against these before it is logged · source config/rules.yaml + offers.yaml, enforced in src/rules.py · rules version 9aa3ddad744a (stored with every decision) · tables below are generated from the config files · all times IST · 23 Oct 2026

Step 1 · Can this customer get anything right now?

Rule	What it says
R1 Consent	Only opted-in customers. /stop works immediately and is checked again at SEND time, per channel. Telegram is the real channel; email, SMS, push and WhatsApp are simulated. /start opts back in. Customers can ask for their data to be erased.
R18 Control group	15% of customers never get offers. Assigned by a hash of a salt + customer ID, so it never changes between decisions. Change the salt only to start a new experiment.
R11 Quiet hours	No SENDING between 22:00 and 08:00 IST. Decisions may run at night; they are planned for the morning window.
R26 Uplift group	Sleeping dogs (reacted badly to our offers) get no offers. Lost causes (gone and not browsing) get free offers only.

Step 2 · Segment (first match wins, in this order)

Each customer also gets a BASE SEGMENT: the same, ignoring the open cart. The open cart decides the offer list (reminder first), but every cap is the stricter of the two, so a VIP with an open cart keeps VIP's limits.

Segment	Condition
CART_ABANDONER	Open cart in the last 48h (checked first, so the reminder-first rule can't be skipped)
NEW	Never ordered, joined ≤ 30 days ago (same meaning as the NEW coupon tier)
NEVER_BOUGHT	Never ordered, joined earlier
DORMANT	No order for max(90 days, 2× their own usual gap)
AT_RISK	3+ orders and 1.5× later than their usual gap, or churn risk ≥ 0.6
POINTS_EXPIRING	300+ points expiring within 30 days
DEAL_SEEKER	3+ orders and coupons on 50%+ of them
VIP	Top 10% by 12-month spend
LOYAL	Buy-anyway score ≥ 0.7
BROWSER	3+ sessions without buying in 14 days
REGULAR	Everyone else

Definitions

Rule	What it says
Usual gap	Average days between a customer's orders. Undefined with 0-1 orders: then DORMANT uses 90 days only, and AT_RISK (3+ orders) doesn't apply. Gap ratio = days since last order / usual gap.
Churn risk	A formula, not a trained model: 0.5 × min(gap ratio / 3, 1) + 0.3 × min(days since last order / 120, 1) + 0.2 × (no order in 90 days). 0 for customers who never ordered.
Buy chance (7 days)	Estimated chance of buying in the next 7 days WITHOUT an offer, from the customer's non-coupon purchase rate, discounted when they are overdue.

Uplift groups (each customer gets one)

Group	Rule of thumb	Effect
Persuadable	Everyone not below	Where offers pay off
Sure thing	Buy chance in 7 days ≥ 30% and coupons on < 30% of orders	Discounts mostly wasted; small, rare coupons only because coupons_for_sure_things = true
Lost cause	No order for 180+ days and no visits in 90 days	Free offers only (R26)
Sleeping dog	2+ bad reactions to our offers in 90 days ('fewer offers', 'not interested' on an offer, /stop). Hiding products doesn't count	No offers (R26)

R2 · Allowed offers per segment, and R4 caps

The fallback picks the allowed offer with the highest expected profit (R30); free nudges count as EP ≥ 0, so they win when no paid offer pays. Ties go to the cheaper offer, then this list's order. Max % applies to store-wide % discounts, Max flat ₹ to flat discounts, and Max cost ₹ to ANY offer's cost (free shipping, points, coupons). Caps are the stricter of segment and base segment.

Segment	Allowed offers (tie-break order; NONE always last)	Max %	Max flat ₹	Max cost ₹	Coupon cap
CART_ABANDONER	CART_REMINDER, CART_ALTERNATIVE, PRODUCT_COUPON, FREE_SHIPPING, FLAT_OFF, BUNDLE_SUGGEST, WISHLIST_ALERT, BUNDLE_DEAL, SPEND_GOAL, NONE	0%	₹150	₹150	tier

Segment	Allowed offers (tie-break order; NONE always last)	Max %	Max flat ₹	Max cost ₹	Coupon cap
NEW	WELCOME, FLAT_OFF, PRODUCT_COUPON, FREE_SHIPPING, WISHLIST_ALERT, BUNDLE_SUGGEST, BUNDLE_DEAL, SPEND_GOAL, NONE	10%	₹100	₹100	tier
NEVER_BOUGHT	WELCOME, PRODUCT_COUPON, FREE_SHIPPING, WISHLIST_ALERT, BUNDLE_SUGGEST, NONE	10%	₹100	₹100	tier
DORMANT	FLAT_OFF, PRODUCT_COUPON, PERCENT_OFF, CATEGORY_DEAL, BONUS_POINTS, POINTS_UPDATE, WISHLIST_ALERT, BUNDLE_SUGGEST, BUNDLE_DEAL, SPEND_GOAL, NONE	20%	₹300	₹300	tier
AT_RISK	CATEGORY_DEAL, PRODUCT_COUPON, REPLENISH_REMINDER, BONUS_POINTS, FLAT_OFF, PERCENT_OFF, FREE_SHIPPING, POINTS_UPDATE, WISHLIST_ALERT, BUNDLE_SUGGEST, BUNDLE_DEAL, SPEND_GOAL, PERK_CHOICE, NONE	15%	₹250	₹250	tier
POINTS_EXPIRING	POINTS_REMINDER, POINTS_UPDATE, BONUS_POINTS, PRODUCT_COUPON, WISHLIST_ALERT, BUNDLE_SUGGEST, NONE	0%	₹0	₹60	5%
DEAL_SEEKER	FLAT_OFF, PRODUCT_COUPON, CATEGORY_DEAL, FREE_SHIPPING, BONUS_POINTS, REPLENISH_REMINDER, REVIEW_REQUEST, WISHLIST_ALERT, BUNDLE_SUGGEST, BUNDLE_DEAL, SPEND_GOAL, PERK_CHOICE, NONE	10%	₹100	₹120	tier
VIP	EARLY_ACCESS, POINTS_UPDATE, REPLENISH_REMINDER, REVIEW_REQUEST, FREE_SHIPPING, BONUS_POINTS, PRODUCT_COUPON, WISHLIST_ALERT, BUNDLE_SUGGEST, PERK_CHOICE, NONE	0%	₹0	₹100	5%
LOYAL	EARLY_ACCESS, POINTS_UPDATE, REPLENISH_REMINDER, POINTS_REMINDER, REVIEW_REQUEST, BONUS_POINTS, PRODUCT_COUPON, WISHLIST_ALERT, BUNDLE_SUGGEST, NONE	0%	₹0	₹100	5%
BROWSER	CATEGORY_DEAL, FLAT_OFF, PRODUCT_COUPON, FREE_SHIPPING, WISHLIST_ALERT, BUNDLE_SUGGEST, BUNDLE_DEAL, SPEND_GOAL, NONE	10%	₹100	₹100	tier
REGULAR	REPLENISH_REMINDER, POINTS_UPDATE, REVIEW_REQUEST, BONUS_POINTS, FREE_SHIPPING, FLAT_OFF, PRODUCT_COUPON, WISHLIST_ALERT, BUNDLE_SUGGEST, BUNDLE_DEAL, SPEND_GOAL, PERK_CHOICE, NONE	5%	₹50	₹100	tier

Step 3 · How big the offer can be

Rule	What it says
R3 Catalog range	Value must be within the offer's own range (e.g. % off 5–20%, flat ₹50–300, product coupon 5–25%).
R4 Segment cap	% or flat ₹ within the segment's maximum; product coupons within the segment's coupon cap.
R5 Margin floor	A discount must leave 10% margin in that category (Mobiles 12%, Electronics 18%, Grocery 15%, Books 25%, Home 35%, Beauty 40%, Fashion 45%). % offers: the % is capped. Flat ₹ off and free shipping: the minimum basket rises until the cost is within the allowed % (₹150 off in Electronics needs a ₹1,880 basket), and the offer is blocked if that exceeds 3× the average order. Free delivery, flat ₹ off and % off only work in categories that keep the floor at that basket (e.g. not Mobiles), and the message names them. Impossible offers are skipped with the reason.
R6 Flat size	Flat ₹ off ≤ 15% of the customer's average order (₹900 if they have no history). Below the offer minimum → skipped, next offer tried.
R21 First order only	WELCOME and the FIRST_ORDER coupon tier are only for customers who have never ordered. WELCOME's cost uses ₹900 as the order value (10% = ₹90).
Welcome scope	Welcome applies only to categories that can afford 10% (Fashion, Home, Beauty, Books).

R19 / R20 · Coupon tiers (product coupons)

A product coupon applies to the one product the customer wants most (cart > wishlist > last viewed > popular in category; never out of stock or sensitive). Frequent buyers always get less, and less often, than occasional, irregular or first-order customers (checked at startup). FIRST_ORDER is not the most generous tier: new accounts are the easiest to fake, so it needs a minimum basket. Goodwill exception: with coupons_for_sure_things on, a customer who buys anyway gets a small coupon when due, although its expected profit is negative.

Tier	Who	Max %	Max ₹	Min basket	At most every
FREQUENT	Buy chance in 7 days ≥ 30%	5%	₹100	-	30 days
OCCASIONAL	Buy chance in 7 days 10–30%	10%	₹200	-	14 days
IRREGULAR	Buy chance in 7 days under 10%	15%	₹400	-	7 days
FIRST_ORDER	Never ordered (segment NEW or NEVER_BOUGHT)	10%	₹200	₹499	7 days

Step 4 · Timing and frequency

Rule	What it says
R8 Active offer	While a customer holds an offer they were told is valid (coupon, free delivery, points, goal...), they get no new offer of any kind until it ends; the daily check skips them. One active paid offer at a time, and never the same offer again while it is still valid.
R9 Frequency	Paid offers: at least 24h apart, at most 3 per 7 days (168h apart, i.e. one a week, after 'fewer offers'). Any message: at least 4h apart, at most 5 per 7 days. Cart reminders skip the gaps (they only matter inside the 48h window) but count towards the weekly limits.
R10 Discount cooldown	No new discount within 14 days of using one.
R12 Reminder first	Anyone with an open cart gets a free cart reminder (or the in-stock alternative) first; paid offers only 24h later. If the reminder doesn't lead to a purchase, the next cart message is a discount (product coupon, ₹ off or free delivery) when one pays, never a second reminder.
R20 Coupon interval	Product coupons at most every 30 / 14 / 7 / 7 days (FREQUENT / OCCASIONAL / IRREGULAR / FIRST_ORDER).

Offer validity and cost

Validity drives R8 and must be stated honestly in the message (R15). Costs are the maximum liability (as if redeemed).

Offer	Valid for	Told to customer	Cost counted by R13
CART_REMINDER	2 days	no (internal)	₹0
CART_ALTERNATIVE	2 days	no (internal)	₹0
POINTS_REMINDER	7 days	no (internal)	₹0
REPLENISH_REMINDER	7 days	no (internal)	₹0
REVIEW_REQUEST	7 days	no (internal)	₹0
POINTS_UPDATE	7 days	no (internal)	₹0
EARLY_ACCESS	7 days	yes	₹0
BONUS_POINTS	7 days	yes	points × ₹0.25
FREE_SHIPPING	5 days	yes	₹60 shipping
FLAT_OFF	5 days	yes	the ₹ value
PERCENT_OFF	5 days	yes	average order × %
CATEGORY_DEAL	5 days	yes	average order × %
PRODUCT_COUPON	7 days	yes	product price × %
WELCOME	7 days	yes	average order × %
BUNDLE_SUGGEST	7 days	no (internal)	₹0
BUNDLE_DEAL	5 days	yes	the ₹ value (only on the add-on item)
WISHLIST_ALERT	7 days	no (internal)	₹0
SPEND_GOAL	7 days	yes	the ₹ value
PERK_CHOICE	5 days	yes	₹60 (either perk)

R30 · Expected profit

EP = (P_with − P_without) × margin − P_with × cost. Every paid offer needs EP > 0, except the goodwill coupon. The AI is given each option's EP and the checker recomputes it. With a minimum order, a customer who buys because of the offer fills the basket up to it, so their margin (and a % discount's cost) uses max(typical order, minimum order).

Rule	What it says
P_without	Buy chance in 7 days without an offer (see Definitions).
P_with	P_without + (1 − P_without) × lift. Lift is the engine's own estimate from observable signals only: discount sensitivity = 0.1 + 0.6 × max(coupon rate, 2 × promo click rate) + 0.2 × churn risk (max 0.6), times discount% / (discount% + 7). Free nudges have fixed lifts when their condition holds.
margin	Average order (₹900 if none) × category margin % (for welcome: the eligible categories' margin).
cost	The offer's cost (table above).

Step 5 · Budget

Rule	What it says
R13 Daily budget	₹20,000 a day across all customers, counted as max liability.
R13 Pacing	Released evenly over 08:00–22:00: by the end of hour h of the window, at most daily × h / 14 may be committed, so an 08:00 rush can't leave evening customers with nothing.
R13 Atomic	The final budget check and the log write happen in one database transaction, so two decisions (even in two programs) can't both spend the last rupees.
Expected cost	Logged with every decision: cost × chance the customer buys while holding the offer. Used to plan the budget; the maximum liability above stays the hard cap.
R31 Per customer	Offer cost given to one customer over 30 days ≤ 10% of the margin they brought in over 12 months (at least ₹100), so one customer can't soak up repeated offers.

Step 6 · Guards

Rule	What it says
R22 No stacking	No % or flat discount while a store promotion already covers that category (one discount per order).
R23 Abuse	No paid offers if return rate ≥ 30% (3+ orders) or 2+ coupon orders returned.
R24 Discount training	If the last 3 orders all used a coupon, no discount now (so customers don't learn to wait for deals). Replaces the planned R14.
R25 Stock	Never promote an out-of-stock product. If the cart item is out of stock, the cart reminder becomes CART_ALTERNATIVE (a similar in-stock product), or nothing.
Coupon codes	Every paid offer gets a unique, single-use code tied to the customer, added at send time. A code used by anyone else, twice, or after expiry is refused and logged as misuse (shared codes are caught).
R28 Feedback	'Not interested' suppresses that offer for 30 days; 'fewer offers' slows the customer down; 'stop' opts out.
R29 Free nudges	Restock reminders (Grocery/Beauty, at the customer's own reorder cycle), review requests and points updates only when their condition holds. They cost nothing, so they are the default when unsure. Bought-together only with a real 'bought with' item in stock; wishlist alerts only for a real price drop or restock; pick-your-perk is not asked again once the customer has picked.

Step 7 · The message

Rule	What it says
R15 Honest message	≤ 300 characters. States the real value and the real expiry. Every number must be a term of the offer or a fact about the customer. The text must match the offer type (a reminder never mentions a discount). No pressure phrases, no invented promises, no links, emails, phone numbers or HTML. Category- and product-limited offers name what they cover.
R16 Sensitive data	Never mentions age, gender, religion, caste, income, health and similar, and never personalises on sensitive product types (e.g. health products). Customer text is sanitised before use.
R17 Valid answer	The AI must return valid JSON with an offer from the catalog.

How an offer is chosen

1. Rules build the allowed offers and value ranges for the customer.
2. The AI picks the offer with the highest expected profit (may be none); the checker re-checks it against every rule, one retry.
3. If the AI fails or breaks the rules twice, the rules-only fallback picks the allowed offer with the highest expected profit (a free nudge when no paid offer pays).
4. R27 exploration: 5% of decisions use a random allowed offer, with its probability logged, so the system can learn which offers work.
5. The decision is logged with the rules version and a planned send time (the customer's usual active hour, 08:00–22:00).
6. Human review is flagged when: the AI failed or broke a rule, an offer costs over ₹300, 90%+ of the daily budget is used, or a reply contains complaint words.
7. Delivery runs in dry run mode with a kill switch, on telegram; consent is checked again at send time and nothing is sent during quiet hours. Paid offers get their single-use code at send time.

Known limits

- Duplicate-account detection needs device/address data the dataset doesn't have (the flag exists, always false).
- Price-drop and back-in-stock alerts need price and stock history; not built.
- One coupon per order at checkout is enforced by the store, not this engine (R22 avoids stacking on store promotions).
- With ~20 classmates the control group is ~3 people: results can't be statistically meaningful at that size.
- India's DPDP Act: consent withdrawal (/stop) and erasure (consent.erase) are built; a privacy notice and retention policy are still needed.